(Open) LLMs
in Behavioral and Social Sciences @ Zurich 2025

Dirk Wulff & Zak Hussain

MAX PLANCK INSTITUTE
FOR HUMAN DEVELOPMENT

UNI
BASEL

Goals

Familiarise you with the workings and applications of open LLMs and how to implement them using the Hugging Face ecosystem

Zak-Hussain / LLM4BeSci_Zurich2025

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Zak-Hussain populate 155e7c9 · 2 hours ago 2 Commits

day_1	populate	2 hours ago
day_2	populate	2 hours ago
.gitignore	Initial commit	4 hours ago
LICENSE.txt	populate	2 hours ago
README.md	populate	2 hours ago
notes.txt	populate	2 hours ago

README License

LLM4BeSci at Uni Zurich, 2025

The course introduces the use of open-source large language models (LLMs) from the Hugging Face ecosystem for research in the behavioral and social sciences.

Materials by Dirk Wulff and Zak Hussain. Taught in-person by Zak Hussain.

Schedule

Day 1

09:30 AM - 09:45 AM: Welcome & Intro

About

The course introduces the use of open-source large language models (LLMs) from the Hugging Face ecosystem for research in the behavioral and social sciences.

Readme
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Activity
0 stars
0 watching
0 forks

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Languages

Jupyter Notebook 100.0%

Schedule

Day 1

09:30 AM - 10:00 AM: Welcome & Intro
10:00 AM - 10:45 AM: Talk: Intro to LLMs
10:45 AM - 11:00 AM: Break
11:00 AM - 11:20 AM: Talk: A gentle intro to Hugging Face and Python
11:20 AM - 11:30 AM: Setup Colab
11:30 AM - 12:00 PM: Exercise: Running pipelines
12:00 PM - 01:00 PM: Lunch
01:00 PM - 01:30 PM: Walkthrough
01:30 PM - 02:00 PM: Talk: Intro to transformers & embeddings
02:00 PM - 02:15 PM: Break
02:15 PM - 03:00 PM: Talk: Intro to transformers & embeddings (continued)
03:00 PM - 04:00 PM: Exercise: Clarifying personality psychology
04:00 PM - 04:15 PM: Break
04:15 PM - 05:00 PM: Walkthrough

Day 2

09:00 AM - 09:30 AM: Recap quiz
09:30 AM - 10:15 AM: Talk: Intro to classification and regression
10:15 AM - 10:30 AM: Break
10:30 AM - 11:30 AM: Exercises: Classifying media bias using text generation and feature extraction
11:30 AM - 12:00 PM: Walkthrough
12:00 PM - 12:30 PM: Discussion: Find applications in small groups

1. What’s your field and research topic?
2. What motivates you to learn more about LLMs?
3. R or Python or __ ?
4. How much experiences do you have with programming and machine learning?

Intro to LLMs

Dirk Wulff & Zak Hussain

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Language models

History

1913
Markov Model
Markov

1956
Rule-based model
Chomsky

1957
“You know a word by the company it keeps”
Firth

1957
Measurement of meaning
Osgood

1986
Word embeddings
Rumelhart et al.

1988
LSA
Deerwester et al.

1996
LSTM
Hochreiter et al.

2001
Direct language model
Bengio et al.

2013
Word2vec
Mikolov et al.

2014
GloVe
Pennington et al.

2017
“Attention is all you need”
Vaswani et al.

1900 1950 2000 2020

Language models

History

1913
Markov Model
Markov (mathematician)

1956
Rule-based model
Chomsky (linguist)

1957
“You know a word by the company it keeps”
Firth (linguist)

1957
Measurement of meaning
Osgood (psychologist)

1986
Word embeddings
Rumelhart et al. (psychologist)

1988
LSA
Deerwester et al. (computer scientist)

1996
LSTM
Hochreiter et al. (computer scientist)

2001
Direct language model
Bengio et al. (CompSci ->)

2013
Word2vec
Mikolov et al.

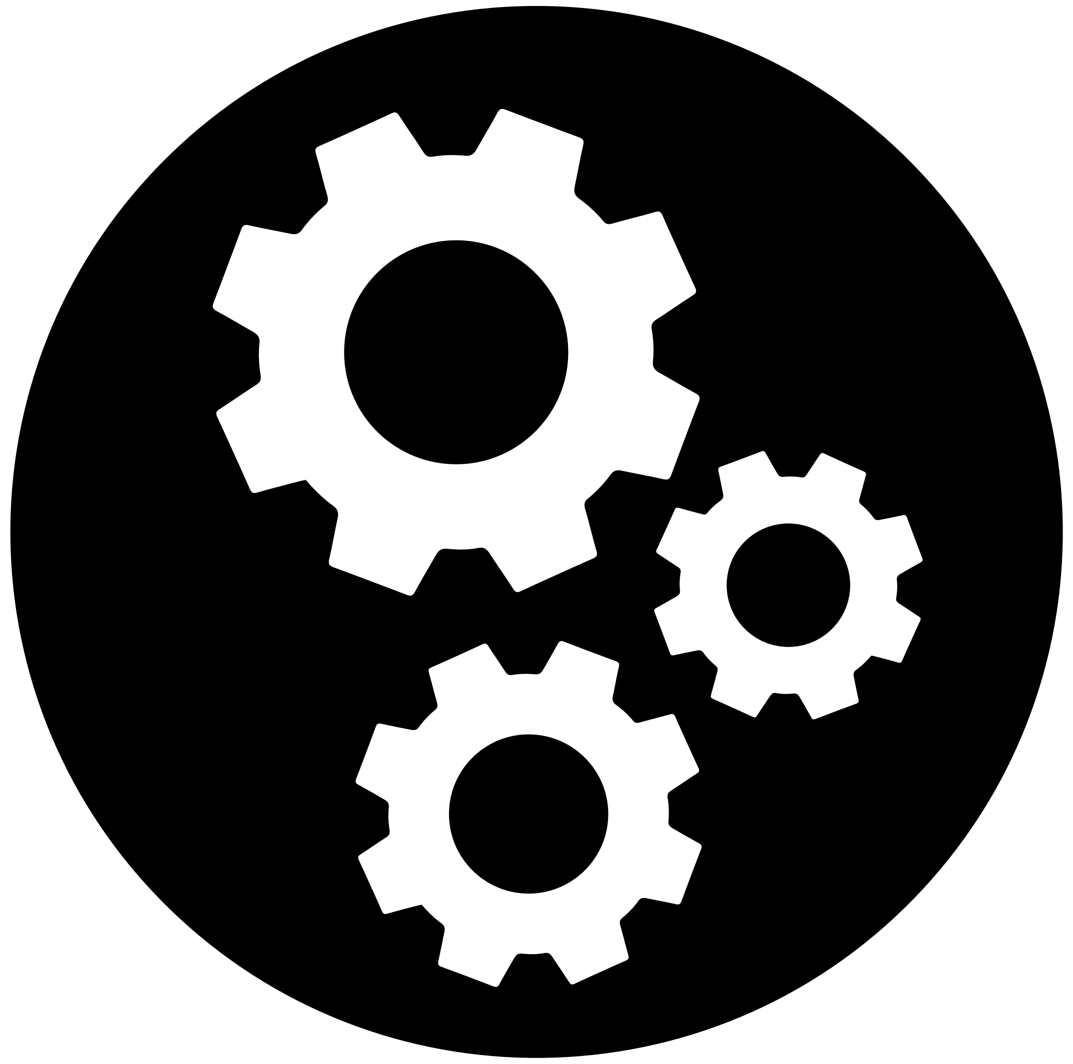
2014
GloVe
Pennington et al.

2017
“Attention is all you need”
Vaswani et al.

1900 1950 2000 2020

LLMs as mechanisms

LLM

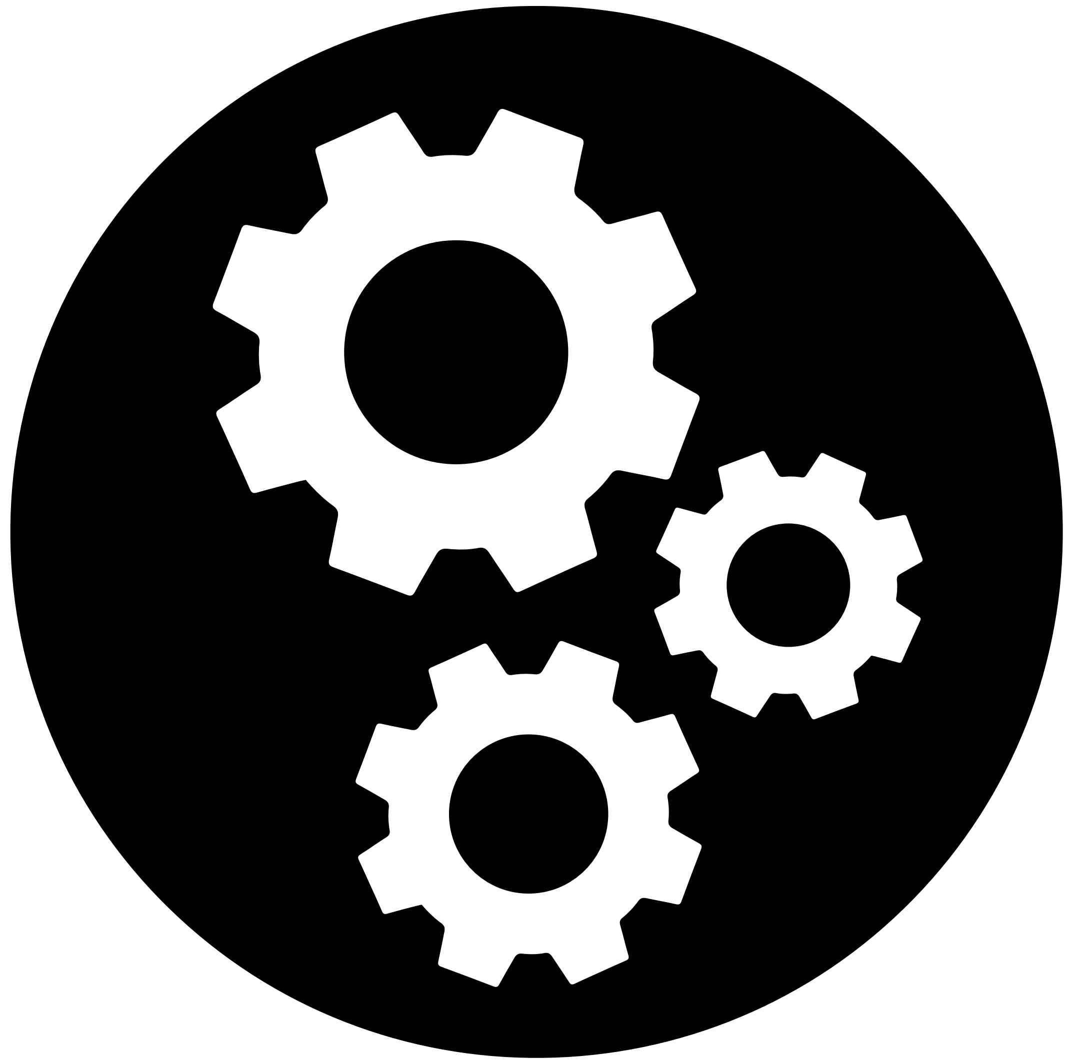


LLMs as mechanisms

Input

“Once upon a time, the” →

LLM

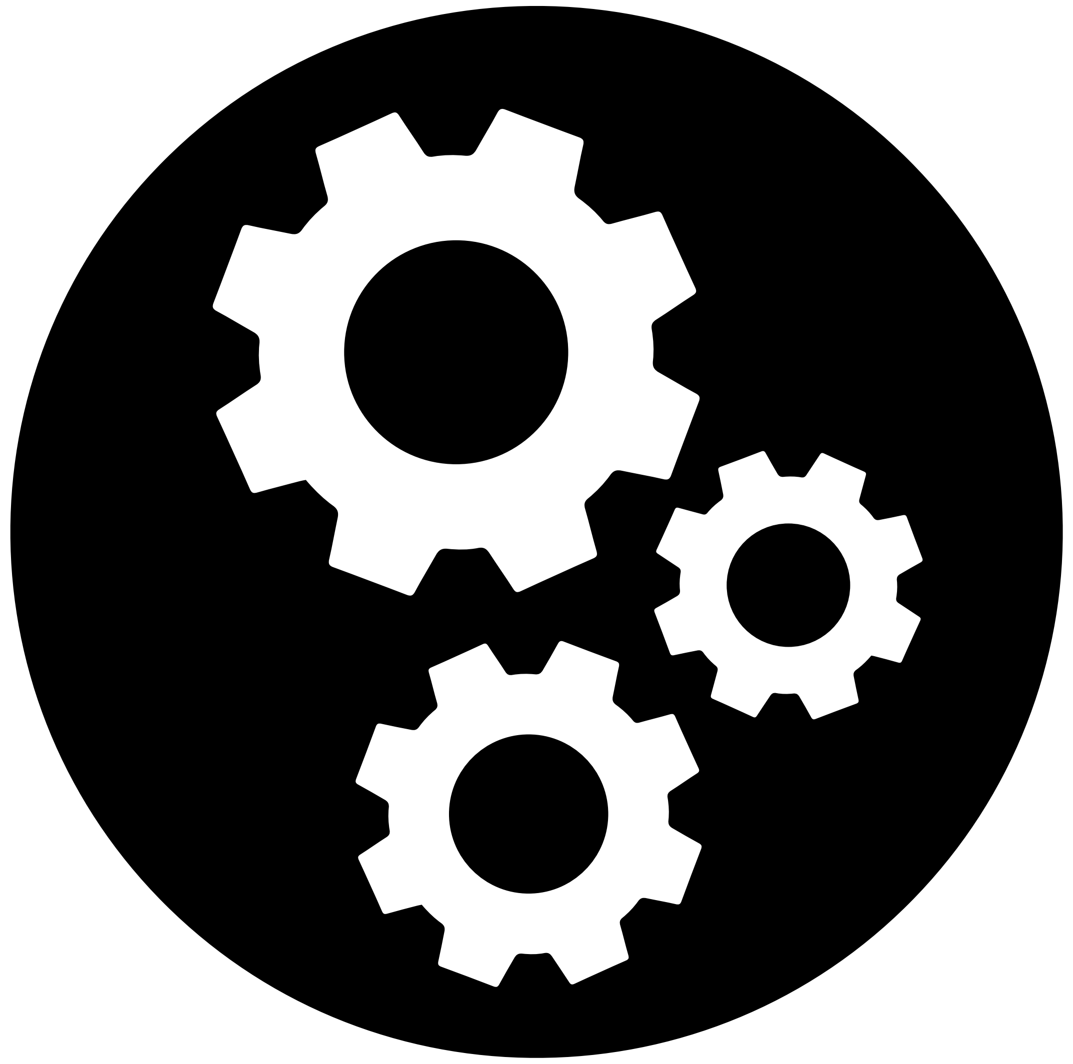


LLMs as mechanisms

Input

"Once upon a time, the" →

LLM



→ **Output**

"dog"

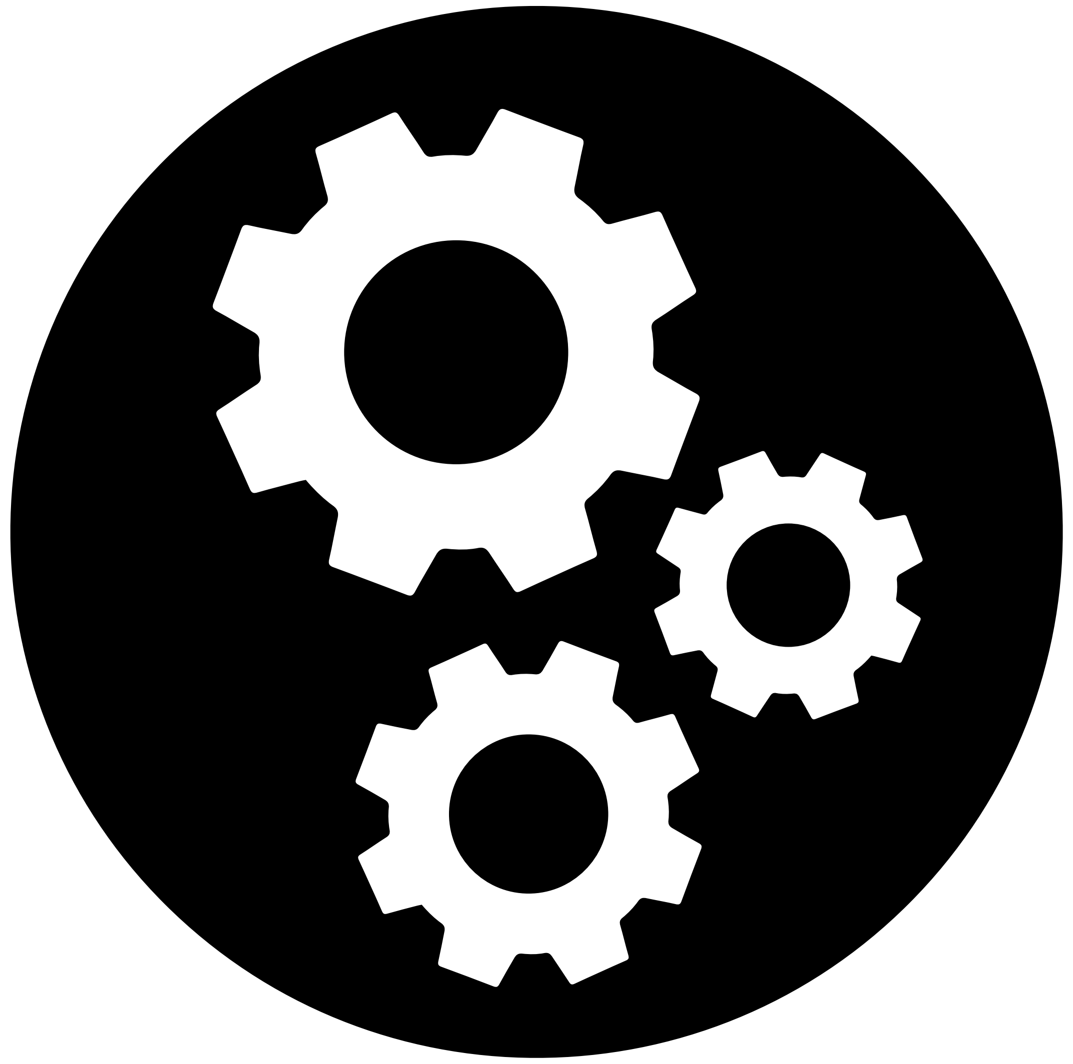
LLMs as mechanisms

Input

“Once upon a time, the”

→

LLM



→

Output

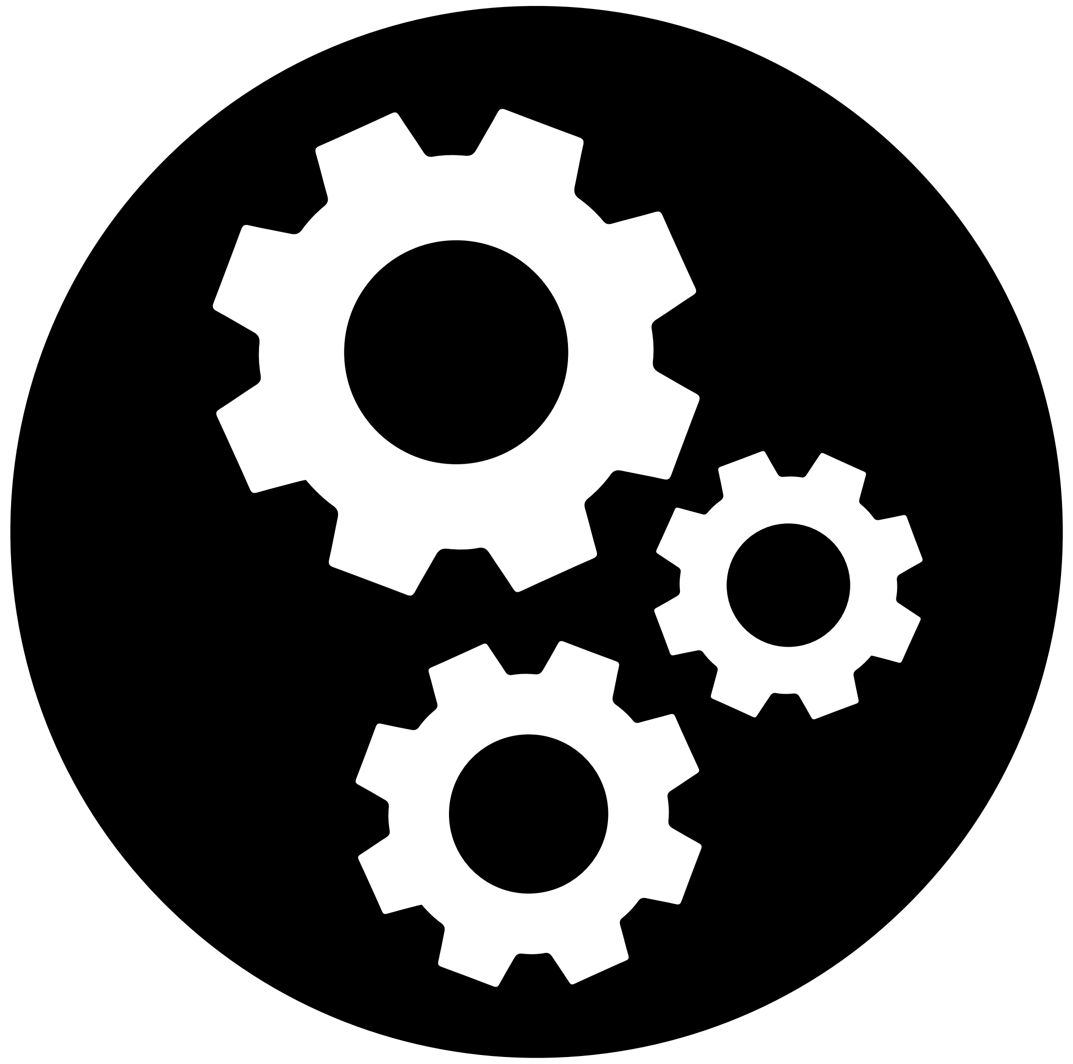
“2”

LLMs as mechanisms

Input

“Once upon a time, the”

LLM



Output

“!”

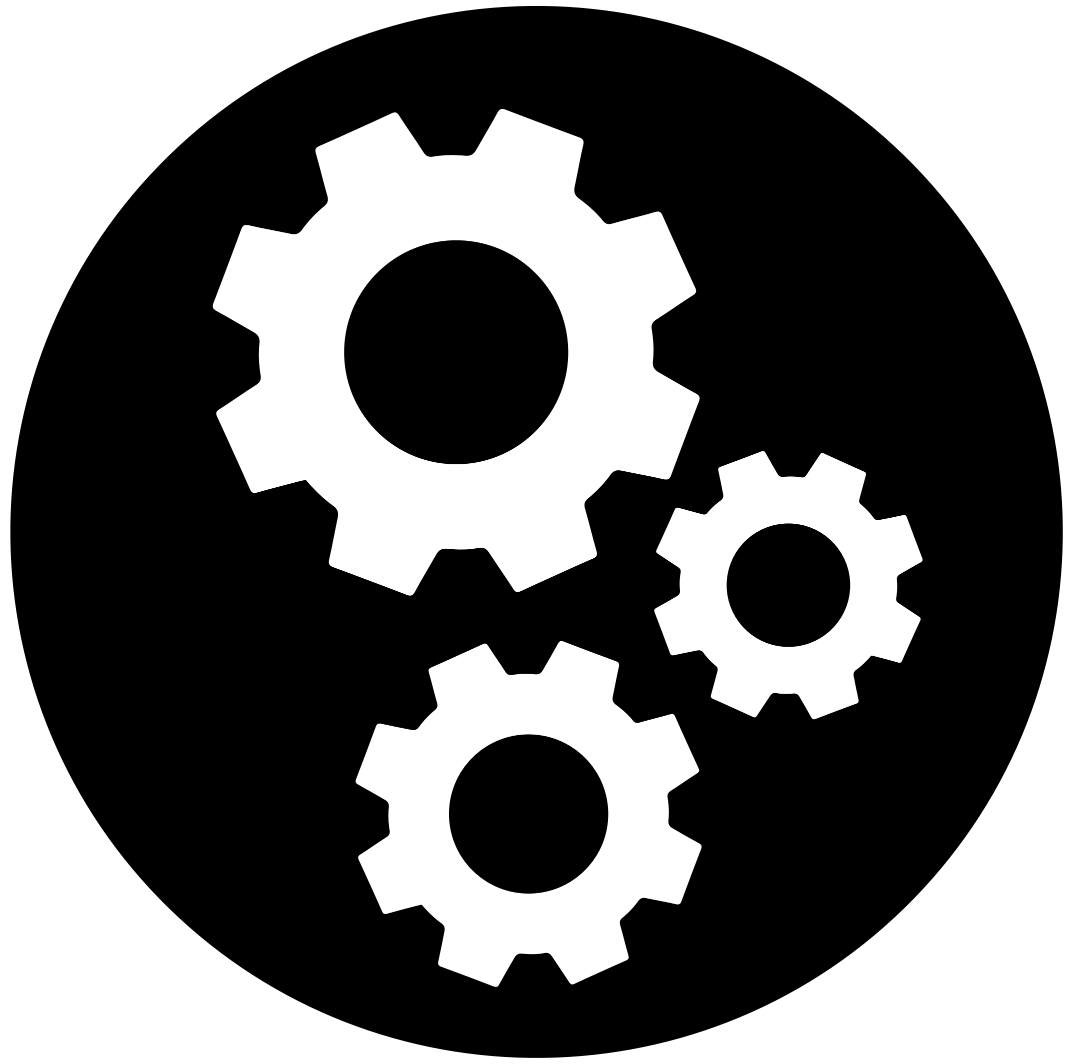
LLMs as mechanisms

Input

“Once upon a time, the”

→

LLM



→

Output

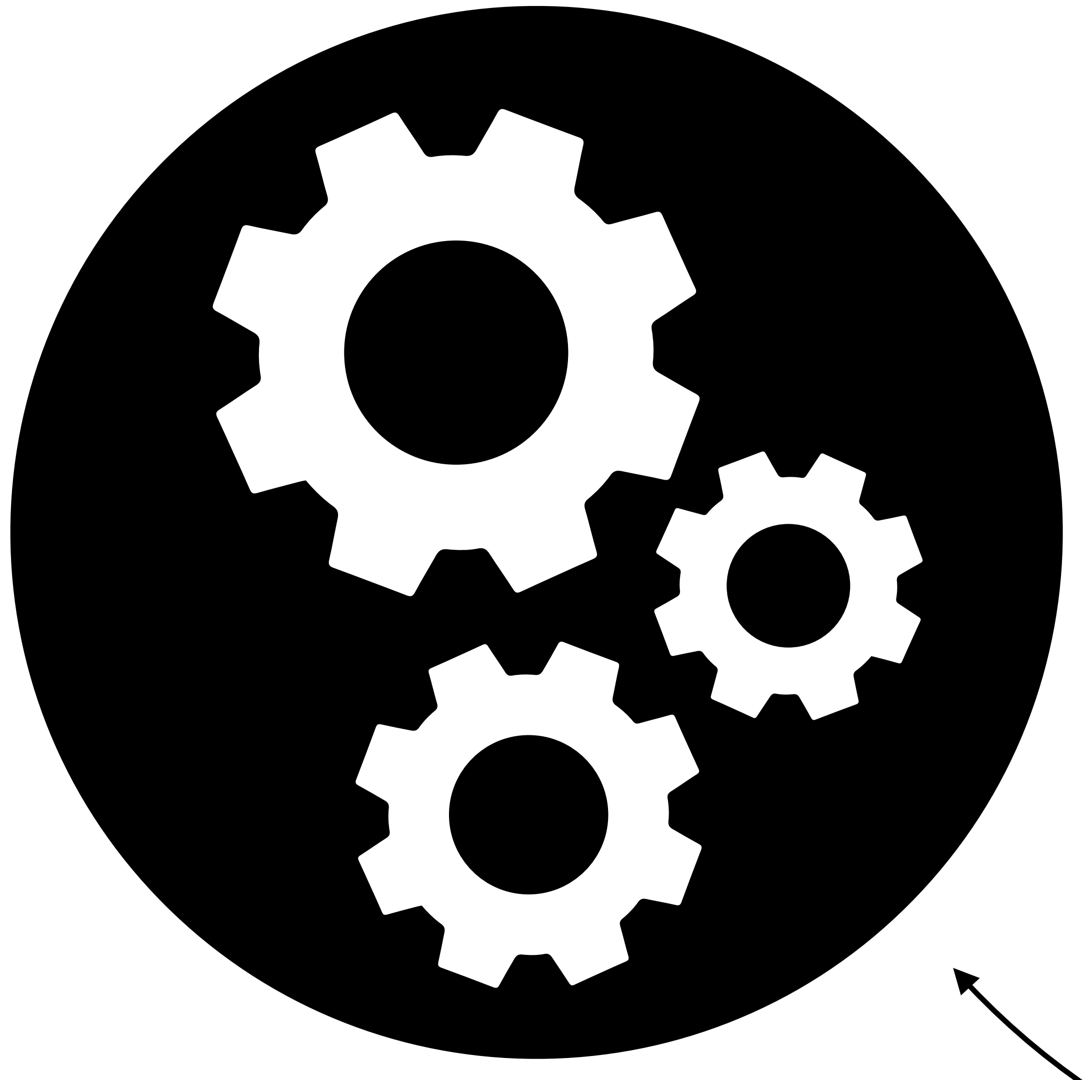
“dog dreamt it was a cat.”

LLMs as trained mechanisms

Input

“Once upon a time, the”

LLM



Output

“dog dreamt it was a cat.”

Feedback

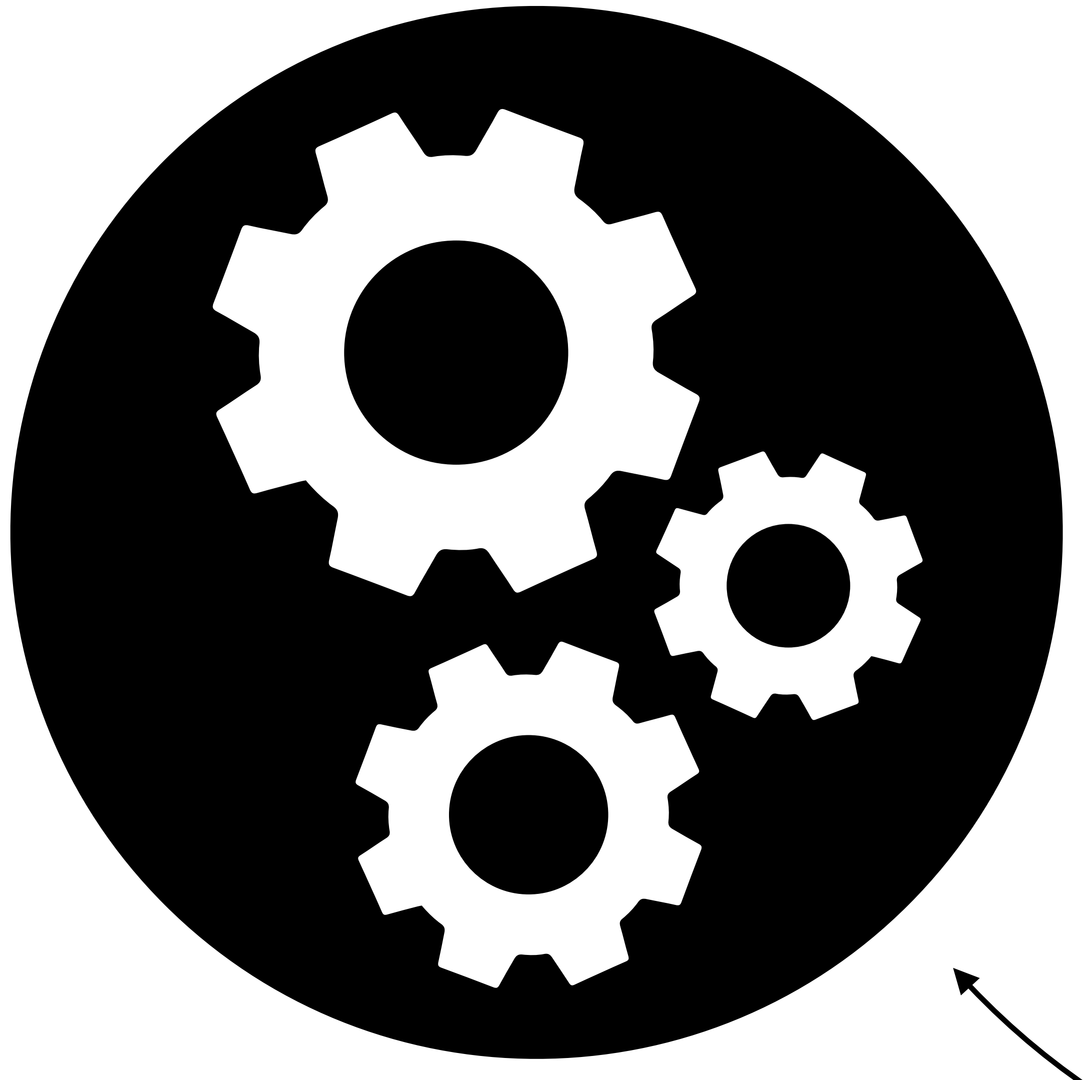
LLMs as trained mechanisms

Input

“Once upon a time, the”

→

LLM



→

Output

“small village of Berlin.”

Feedback

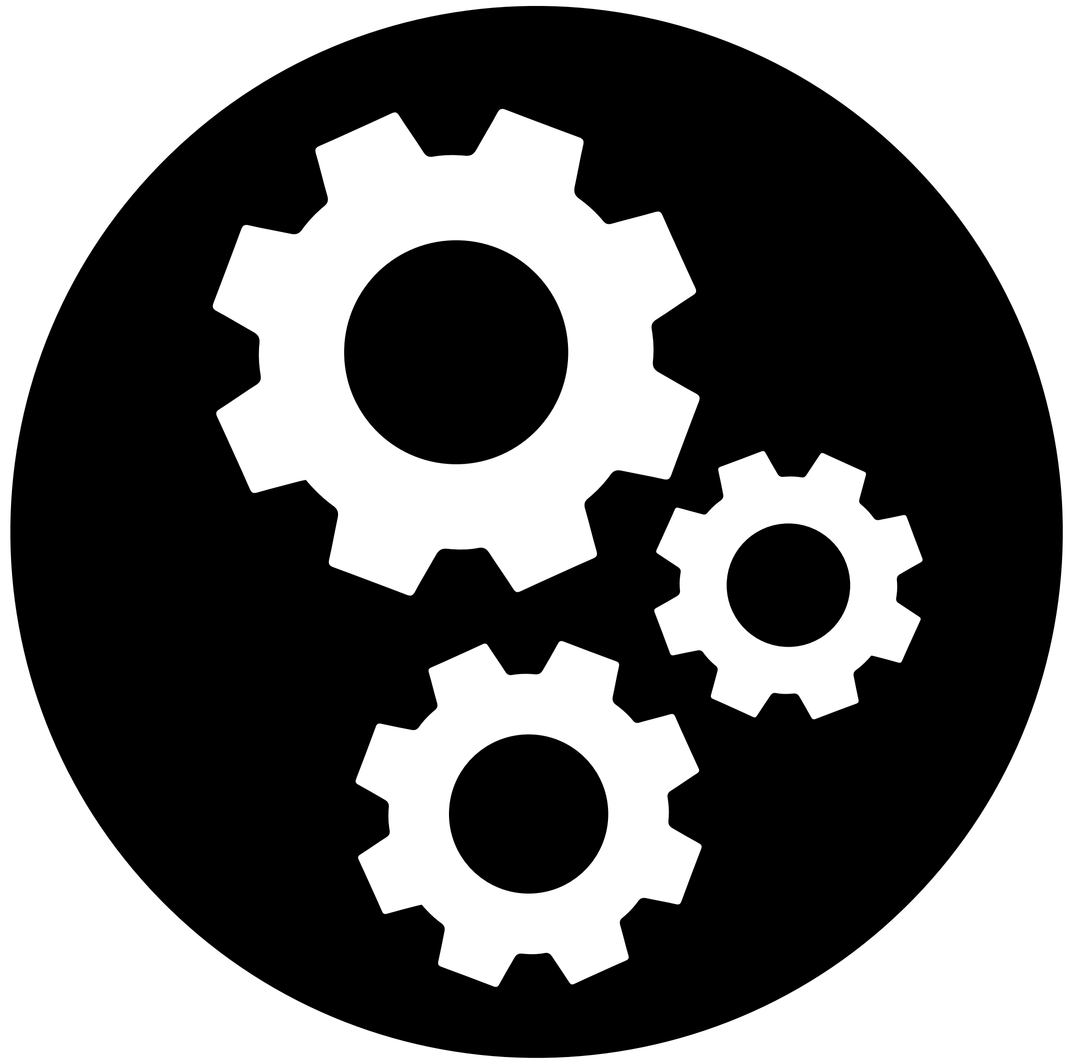
LLMs as trained mechanisms

Input

"Once upon a time, the"

→

LLM



→

Output

"small village of Elmswood…"

Once upon a time, the

Once upon a time, the small village of Elmswood was nestled in a lush valley surrounded by towering mountains. The villagers lived peacefully, their days marked by the rhythms of nature and the changing seasons. However, everything changed when a mysterious old man arrived, carrying with him a locked chest that was said to contain a secret capable of altering the course of history. Intrigued by the stranger and his enigmatic treasure, the people of Elmswood soon found themselves on the brink of an adventure that would bind them together in ways they could never have imagined.

Phi-3-mini-4K-Instruct and Phi-3-mini-128K-Instruct were trained over 7 days on 3.3T tokens using 512 H100-80G GPUs for each model. They followed advanced fine-tuning techniques to align with human preferences and safety standards.

The pre-training process followed two distinct and consecutive stages:

- *In the first stage*, the models were primarily exposed to a vast collection of web sources. This data helped the models develop general knowledge and language comprehension.
- *In the second stage*, the models were fine-tuned with a more rigorously selected subset of web data from the first phase, combined with additional synthetic data, to improve their logical reasoning and specialized abilities.

After these 2 stages, the models underwent *additional training*, which included supervised instruction fine-tuning and preference tuning, to enhance their stability and security.

The training dataset, made of 3.3 trillion tokens, is a meticulously curated mix of quality-filtered public documents, select educational materials, code, and newly generated synthetic data generated by LLMs. Specifically, the team filtered the web data to encompass the appropriate degree of *knowledge* and retained a greater number of web pages that may enhance the models' reasoning abilities. Instead of indiscriminately feeding vast amounts of data into the training model, the emphasis was placed on enhancing its reasoning capabilities, rather than one that merely has a vast repository of information.

Phi-3

LLM training = Pretraining + Fine-tuning

Trillions of tokens
Millions of power consumption
Uses masked/next token prediction

Hand-selected/crafted texts
Quality input-output pairs
Human feedback

Masked/next token prediction

"**Once upon a time**" is a stock phrase used to introduce a narrative of past events, typically in fairy tales and folk tales. It has been used in some form since at least 1380 (according to the *Oxford English Dictionary*) in storytelling in the English language and has started many narratives since 1600. These stories sometimes end with "and they all lived happily ever after", or, originally, "happily until their deaths".

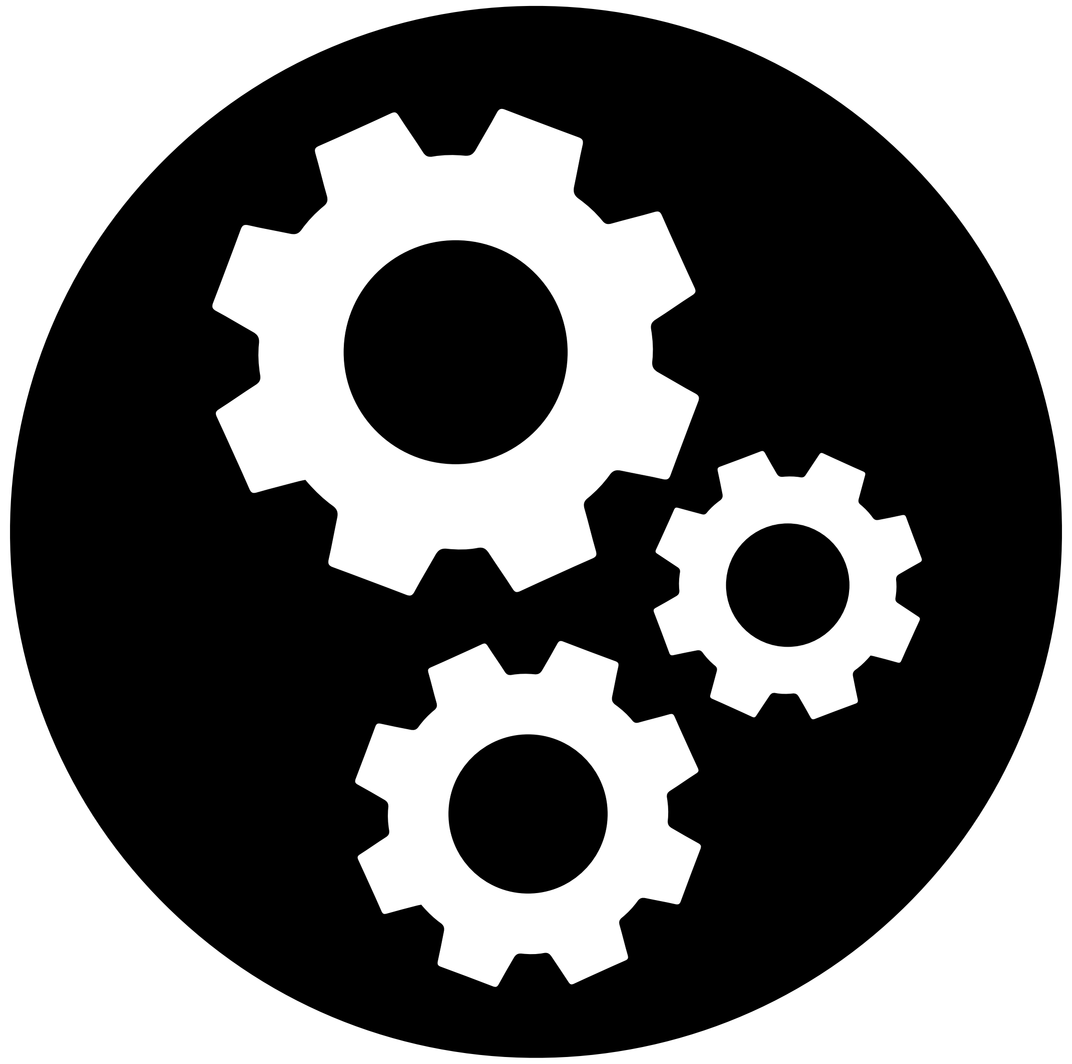
The phrase is common in fairy tales for younger children. It was used in the original translations of the stories of Charles Perrault as a translation for the French "*il était une fois*", of Hans Christian Andersen as a translation for the Danish "*der var engang*" (literally "there was once"), the Brothers Grimm as a translation for the German "*es war einmal*" (literally "it was once") and Joseph Jacobs in English translations and fairy tales.

In *More English Fairy Tales*, Joseph Jacobs notes that:

> "The opening formula are varied enough, but none of them has much play of fancy. 'Once upon a time and a very good time it was, though it wasn't in my time nor in your time nor in any one else's time.' is effective enough for a fairy epoch, and is common, according to Mayhew (London Labour, III), among tramps."[1]

https://en.wikipedia.org/wiki/Once_upon_a_time

LLM



Masked/next token prediction

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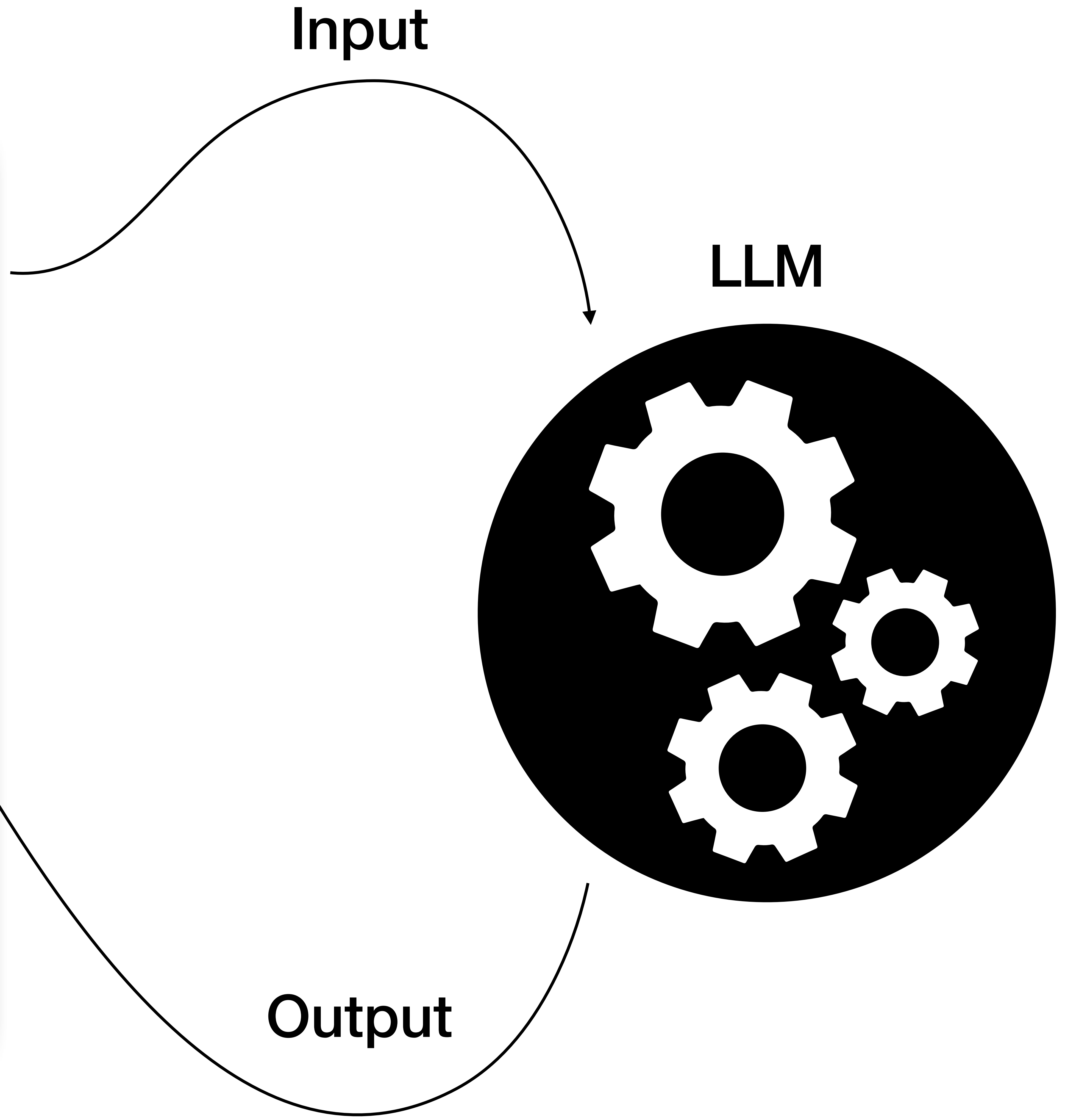

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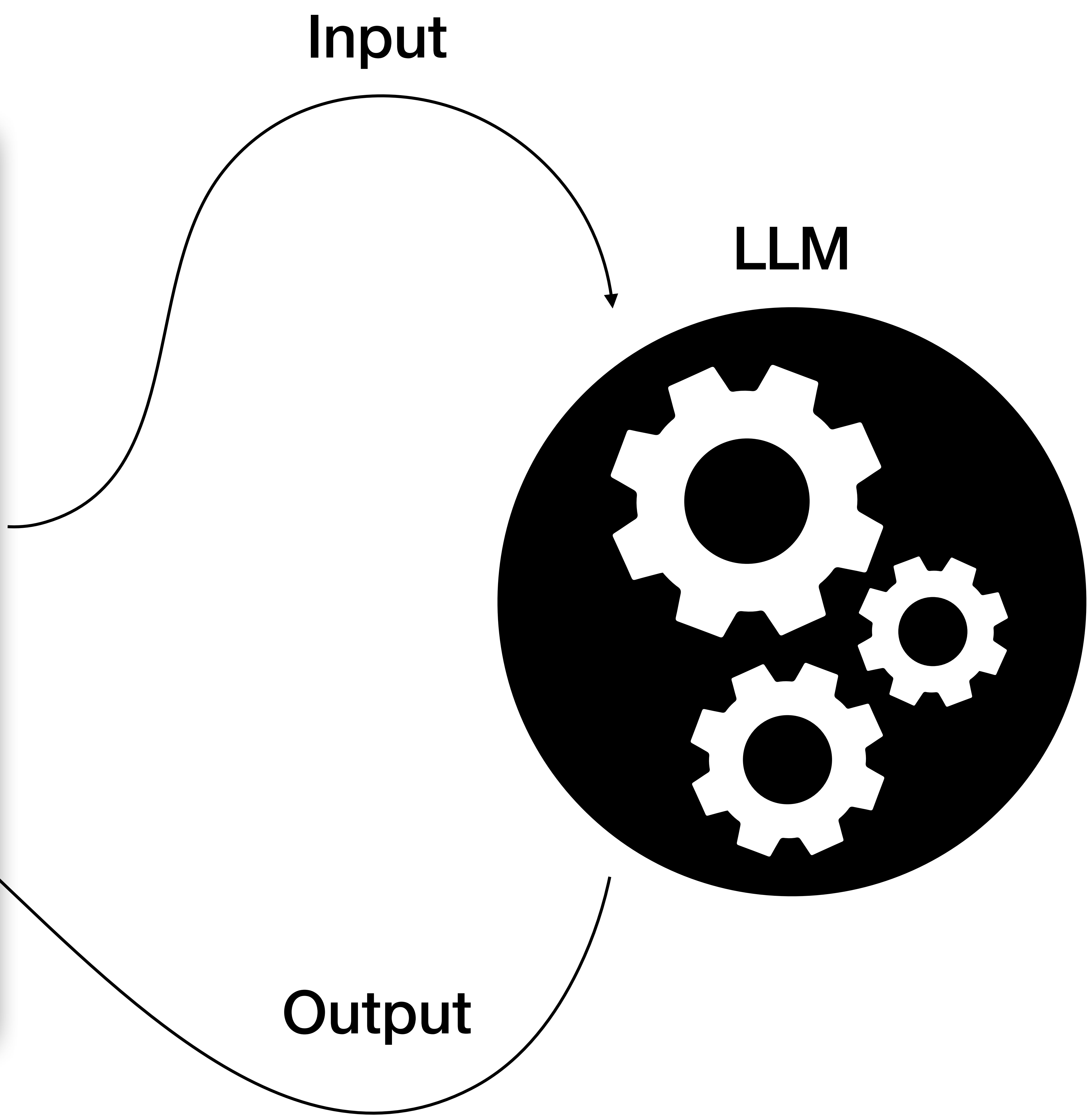

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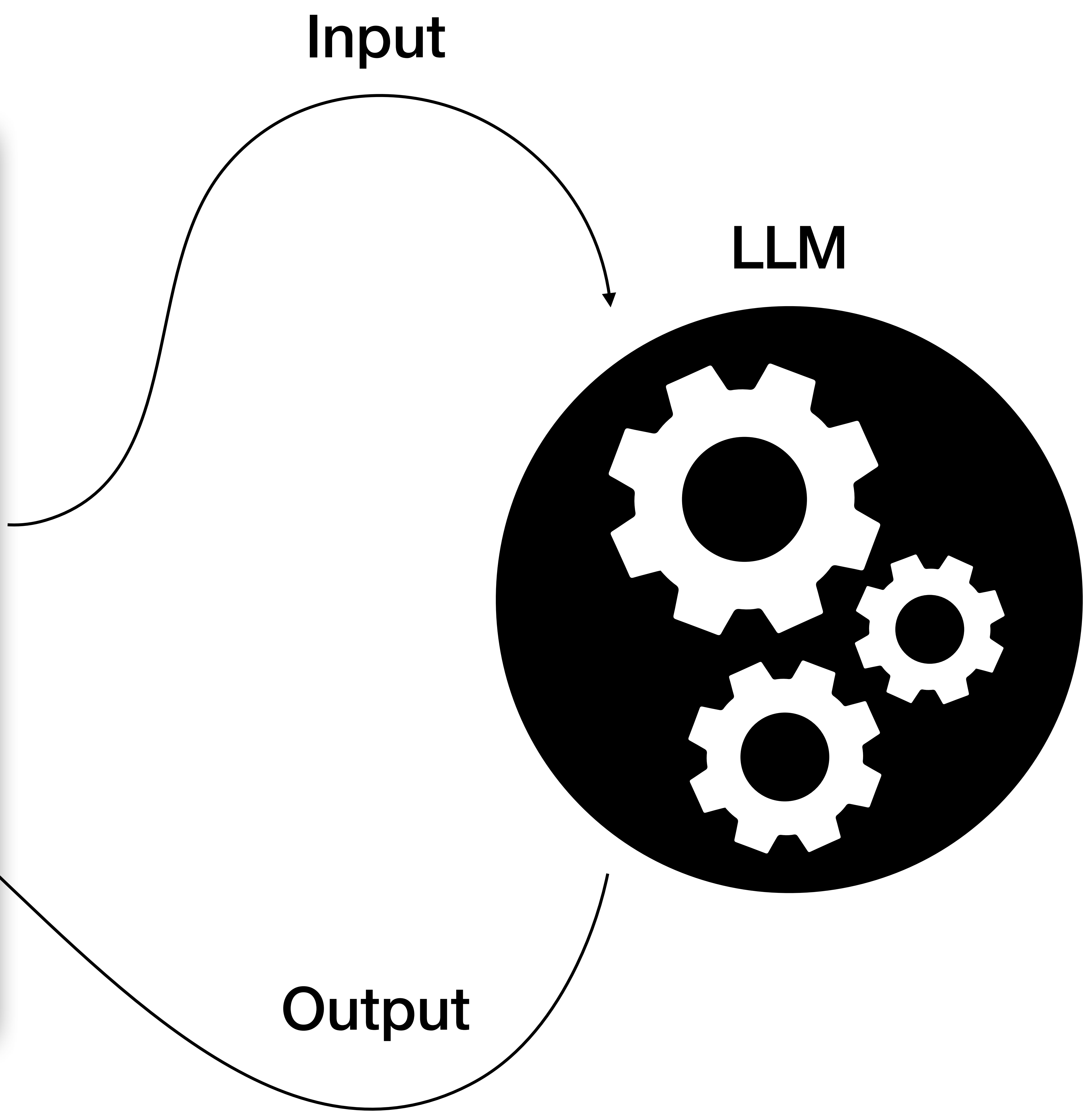

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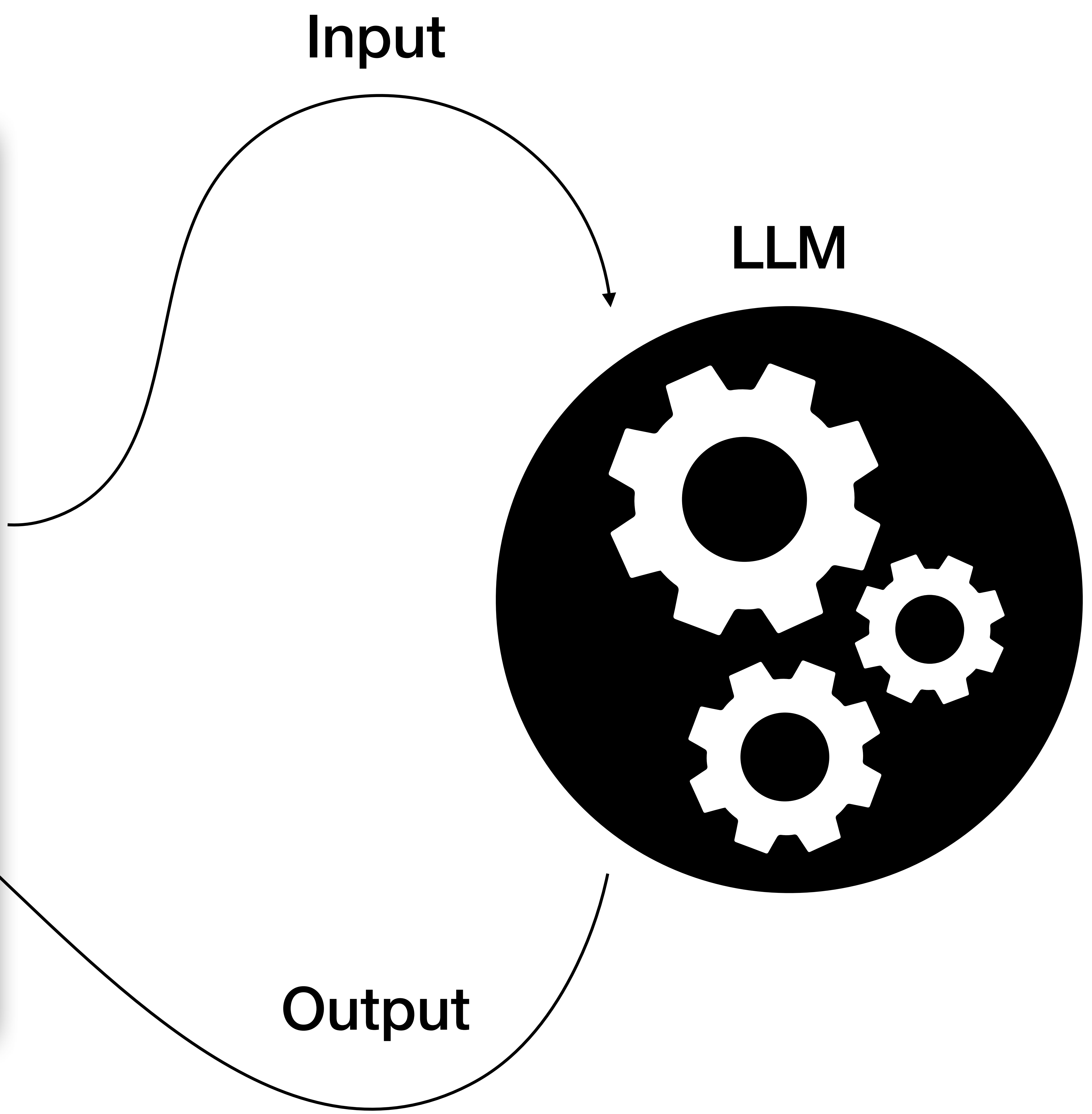

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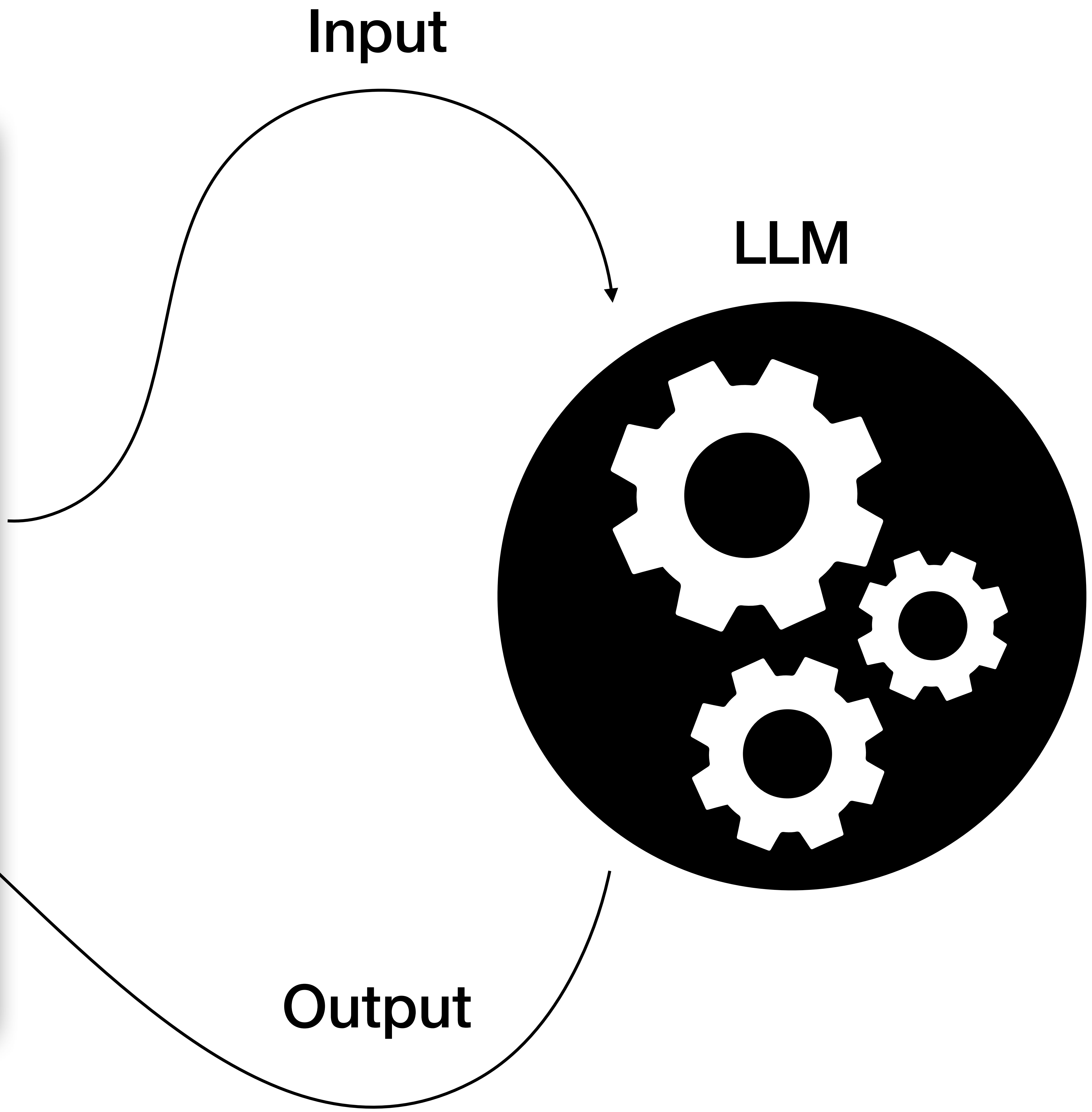
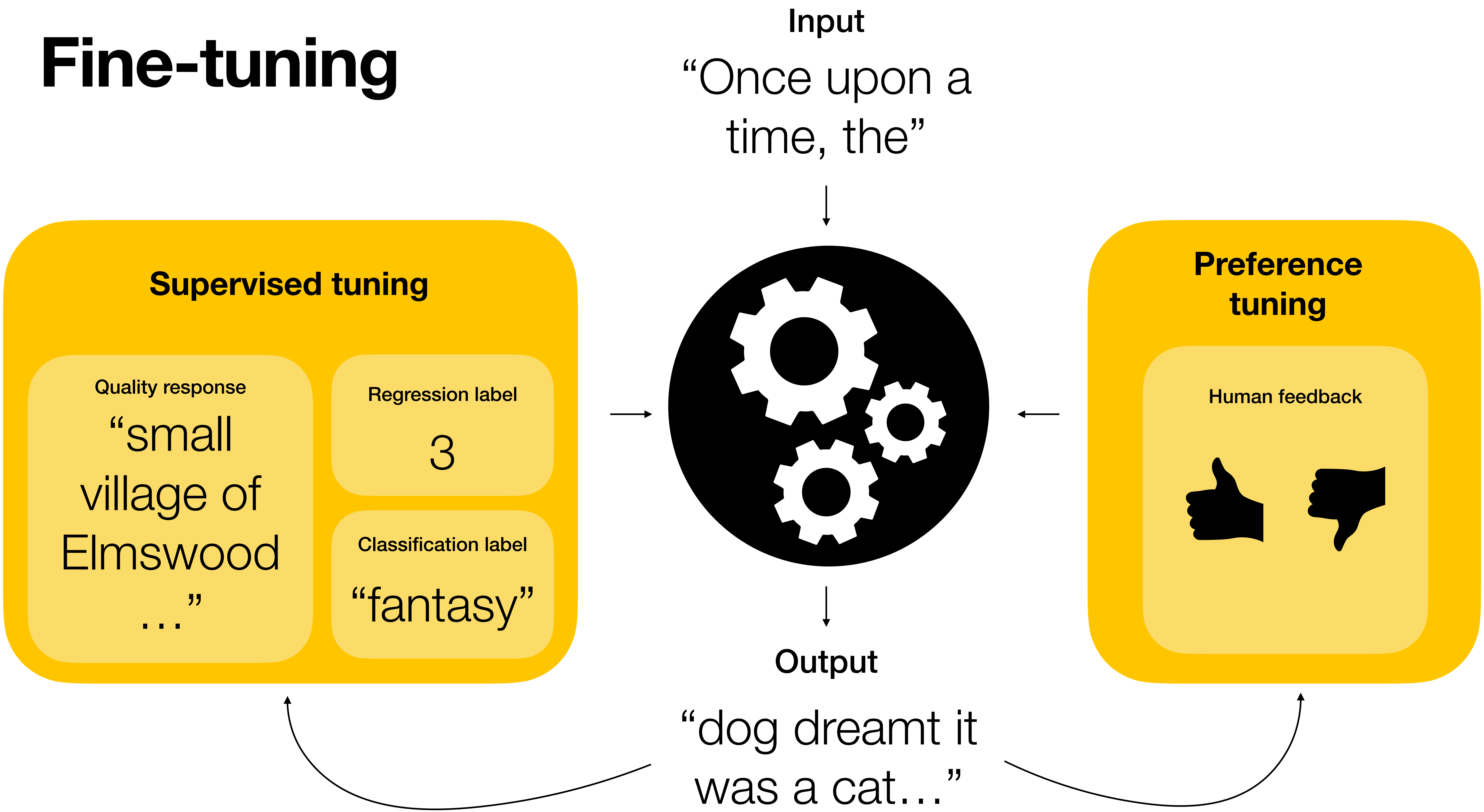


Fine-tuning



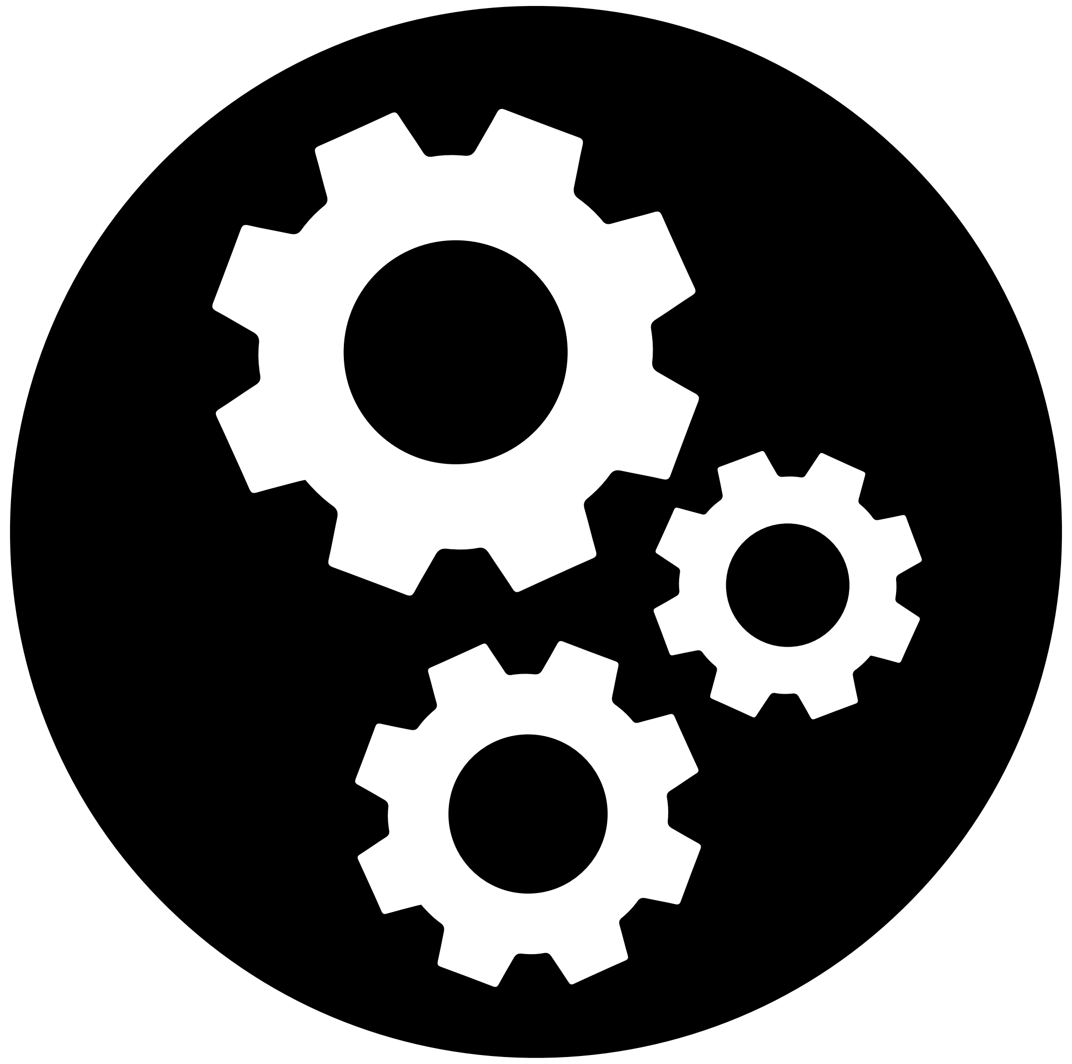
Input
“Once upon a time, the”
Supervised tuning
Quality response
“small village of Elmswood …”
Regression label
3
Classification label
“fantasy”
Preference tuning
Human feedback
Output
“dog dreamt it was a cat…”

Two major applications

Input

“Once upon a time, the”

LLM



Text generation

“small village of Elmswood…”

Feature extraction

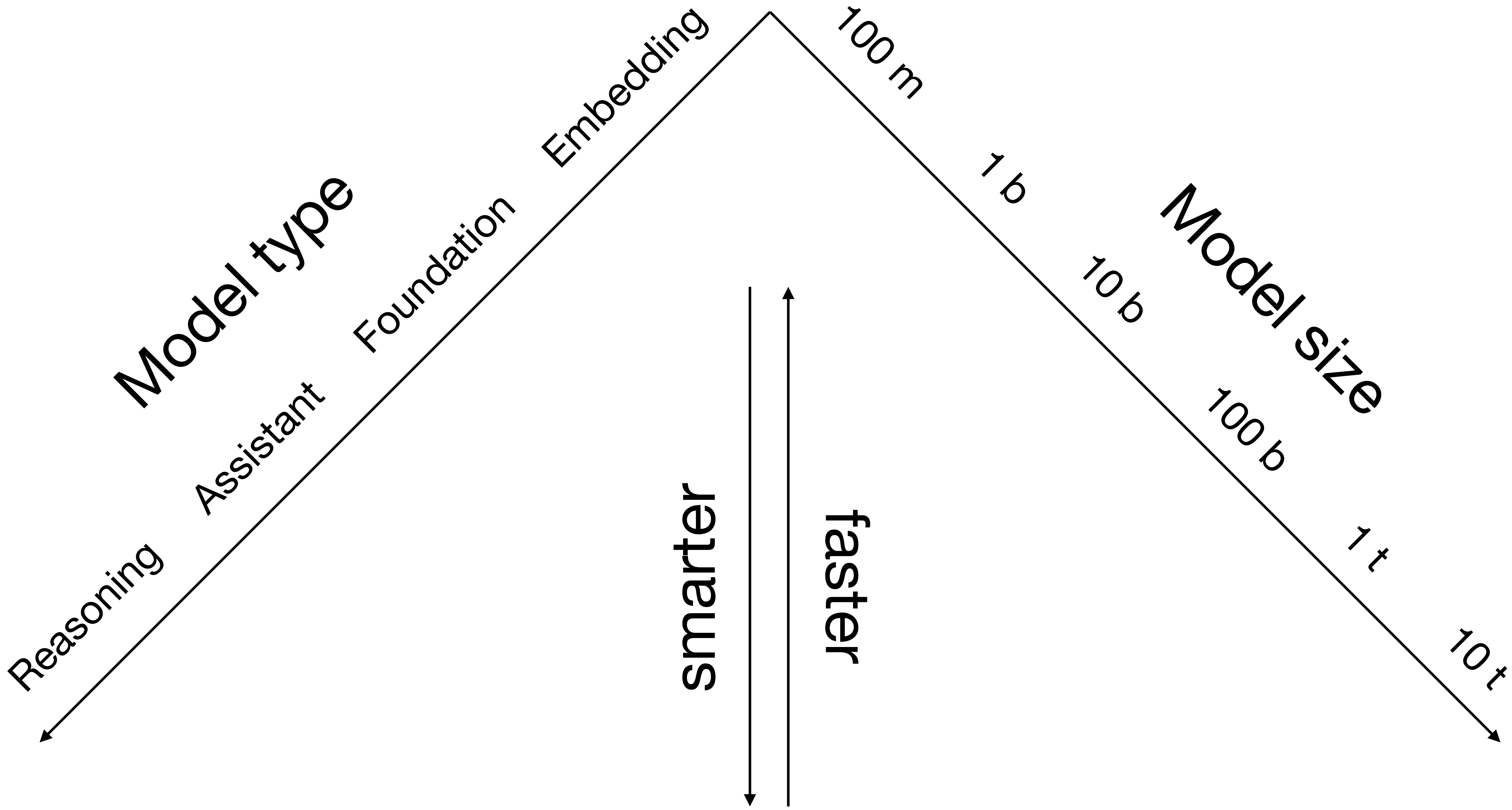
-.23, 1.23, .24, -.12, .34, .32, …

Closed vs. Open

Closed	Open
Public use	Reproducibility
Performance	Data security
Safety	Interpretablity


Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Posts
Docs
Enterprise
Pricing
Tasks
Libraries
Datasets
Languages
Licenses
Other
Filter Tasks by name
Multimodal
Audio-Text-to-Text
Image-Text-to-Text
Visual Question Answering
Document Question Answering
Video-Text-to-Text
Visual Document Retrieval
Any-to-Any
Computer Vision
Depth Estimation
Image Classification
Object Detection
Image Segmentation
Text-to-Image
Image-to-Text
Image-to-Image
Image-to-Video
Unconditional Image Generation
Video Classification
Text-to-Video
Zero-Shot Image Classification
Mask Generation
Zero-Shot Object Detection
Text-to-3D
Image-to-3D
Image Feature Extraction
Keypoint Detection
Natural Language Processing
Text Classification
Token Classification
Table Question Answering
Question Answering
Zero-Shot Classification
Translation
Summarization
Feature Extraction
Models 1,680,396
Filter by name
Full-text search
Sort: Trending
nvidia/parakeet-tdt-0.6b-v2
Automatic Speech Recognition • Updated 10 days ago • 72.7k • 691
ACE-Step/ACE-Step-v1-3.5B
Text-to-Audio • Updated 1 day ago • 310
nari-labs/Dia-1.6B
Text-to-Speech • Updated 5 days ago • 148k • • 2.04k
Lightricks/LTX-Video
Text-to-Video • Updated 5 days ago • 214k • • 1.38k
JetBrains/Mellum-4b-base
Text Generation • Updated 4 days ago • 2.81k • 311
Qwen/Qwen3-235B-A22B
Text Generation • Updated 10 days ago • 82.4k • • 764
deepseek-ai/DeepSeek-Prover-V2-671B
Text Generation • Updated 11 days ago • 7.38k • • 750
lodestones/Chroma
Text-to-Image • Updated 1 day ago • 401
black-forest-labs/FLUX.1-dev
Text-to-Image • Updated Aug 16, 2024 • 2.66M • • 10.1k
cognition-ai/Kevin-32B
Updated 5 days ago • 342 • 93
Qwen/Qwen3-30B-A3B
Text Generation • Updated 11 days ago • 150k • • 514
microsoft/Phi-4-reasoning-plus
Text Generation • Updated 3 days ago • 10.8k • 240
tencent/HunyuanCustom
Image-to-Video • Updated 2 days ago • 77
Tesslate/UIGEN-T2-7B-Q8_0-GGUF
Text Generation • Updated 5 days ago • 2.59k • 115
hexgrad/Kokoro-82M
Text-to-Speech • Updated about 1 month ago • 1.88M • • 4.27k
fdtn-ai/Foundation-Sec-8B
Text Generation • Updated 10 days ago • 21.5k • 145
ServiceNow-AI/Apriel-Nemotron-15b-Thinker
Updated 4 days ago • 784 • 66
Qwen/Qwen3-8B
Text Generation • Updated 12 days ago • 394k • 262

Distinguishing LLMs



Project	Availability						Documentation						Access	
(maker, bases, URL)	Open code	LLM data	LLM weights	RL data	RL weights	License	Code	Architecture	Preprint	Paper	Modelcard	Datasheet	Package	API
OLMo 7B Instruct	✔	✔	✔	✔	✔	✔	✔	✔	✔	✘	✔	✔	✔	~
Ai2	LLM base: OLMo 7B			RL base: OpenInstruct										12.5
BLOOMZ	✔	✔	✔	✔	~	~	✔	✔	✔	✔	✔	✔	✘	✔
bigscience-workshop	LLM base: BLOOMZ, mT0			RL base: xP3										12.0
AmberChat	✔	✔	✔	✔	✔	✔	~	~	✔	✘	~	~	✘	✔
LLM360	LLM base: Amber			RL base: ShareGPT + Evol-Instruct (sy…										10.0
Open Assistant	✔	✔	✔	✔	✘	✔	✔	✔	~	✘	✘	✘	✔	✔
LAION-AI	LLM base: Pythia 12B			RL base: OpenAssistant Conversations										9.5
…														
Command R+	✘	✘	✘	✔	✔	~	✘	✘	✘	✘	~	✘	✘	✘
Cohere AI	LLM base:			RL base: Aya Collection										3.0
LLaMA2 Chat	✘	✘	~	✘	~	✘	✘	~	~	✘	~	✘	✘	~
Facebook Research	LLM base: LLaMA2			RL base: Meta, StackExchange, Anthropic										3.0
Nanbeige2-Chat	✔	✘	✘	✘	✔	~	✘	✘	✘	✘	✘	✘	✘	~
Nanbeige LLM lab	LLM base: Unknown			RL base: Unknown										3.0
Llama 3 Instruct	✘	✘	~	✘	~	✘	✘	~	✘	✘	~	✘	✘	~
Facebook Research	LLM base: Meta Llama 3			RL base: Meta, undocumented										2.5
Solar 70B	✘	✘	~	✘	~	✘	✘	✘	✘	✘	~	✘	✘	~
Upstage AI	LLM base: LLaMA2			RL base: Orca-style, Alpaca-style										2.0
Xwin-LM	✘	✘	~	✘	✘	✘	✘	✘	✘	✘	✘	✘	✘	~
Xwin-LM	LLM base: LLaMA2			RL base: unknown										1.0
ChatGPT	✘	✘	✘	✘	✘	✘	✘	✘	~	✘	✘	✘	✘	✘
OpenAI	LLM base: GPT 3.5			RL base: Instruct-GPT										0.5

see also https://osai-index.eu/the-index

Massive Text Embedding Benchmark (MTEB) Leaderboard. To submit, refer to the MTEB GitHub repository 🤗 Refer to the MTEB paper for details on metrics, tasks and models.

Overall | Bitext Mining | Classification | Clustering | Pair Classification | Reranking | Retrieval | STS | Summarization | Retrieval w/Instructions

English | Chinese | French | Polish

Overall MTEB English leaderboard 🔮

- **Metric:** Various, refer to task tabs
- **Languages:** English

Rank	Model	Model Size (Million Parameters)	Memory Usage (GB, fp32)	Embedding Dimensions	Max Tokens	Average (56 datasets)	Classification Average (12 datasets)	Clustering Average (11 datasets)
1	NV-Embed-v1					69.32	87.35	52.8
2	voyage-large-2-instruct			1024	16000	68.28	81.49	53.35
3	SFR-Embedding-Mistral	7111	26.49	4096	32768	67.56	78.33	51.67
4	gte-Qwen1.5-7B-instruct	7099	26.45	4096	32768	67.34	79.6	55.83
5	voyage-lite-02-instruct	1220	4.54	1024	4000	67.13	79.25	52.42
6	GritLM-7B	7242	26.98	4096	32768	66.76	79.46	50.61
7	e5-mistral-7b-instruct	7111	26.49	4096	32768	66.63	78.47	50.26
8	google-gecko.text-embedding-p	1200	4.47	768	2048	66.31	81.17	47.48
9	SE_v1					65.66	76.8	47.38
10	GritLM-8x7B	46703	173.98	4096	32768	65.66	78.53	50.14

🏆 LMSYS Chatbot Arena Leaderboard

| Vote | Blog | GitHub | Paper | Dataset | Twitter | Discord |

LMSYS Chatbot Arena is a crowdsourced open platform for LLM evals. We've collected over **1,000,000** human pairwise comparisons to rank LLMs with the Bradley-Terry model and display the model ratings in Elo-scale. You can find more details in our paper.

Arena Full Leaderboard

Total #models: **99**. Total #votes: **1,170,955**. Last updated: 2024-05-20.

📣 **NEW!** View leaderboard for different categories (e.g., coding, long user query)! This is still in preview and subject to change.

Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License
1	1	Gemini-Exp-1206	1374	+4/-5	20227	Google	Proprietary
1	1	ChatGPT-4o-latest (2024-11-20)	1365	+4/-3	33383	OpenAI	Proprietary
1	4	Gemini-2.0-Flash-Thinking-Exp-1219	1364	+5/-6	15728	Google	Proprietary
2	4	Gemini-2.0-Flash-Exp	1357	+6/-4	19030	Google	Proprietary
3	1	o1-2024-12-17	1351	+7/-7	7289	OpenAI	Proprietary
6	4	o1-preview	1335	+4/-4	33194	OpenAI	Proprietary
7	7	DeepSeek-V3	1319	+6/-6	10510	DeepSeek	DeepSeek
7	10	Step-2-16K-Exp	1305	+8/-9	3374	StepFun	Proprietary

AI Model Benchmarks Nov 2025

20 benchmarks - the world's most-followed benchmarks, curated by AI Explained, author of SimpleBench

Independently-run benchmarks by Epoch, Scale and others, so may not match self-reported scores by AI orgs.

https://lmcouncil.ai/benchmarks

Compare Models

Model 1: Select...

Model 2: Select...

Compare

Search: Filter...

Humanity's Last Exam

2,500 of the toughest, subject-diverse, multi-modal questions designed to test for both depth of reasoning and breadth of knowledge. Created in partnership with the Center for AI Safety, HLE includes questions across mathematics, humanities, and natural sciences from nearly 1,000 expert contributors.

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	MODELS (NO TOOLS)	SCORE
1	GPT-5 (August '25)	25.32% ±1.70
2	Kimi K2 Thinking	23.9% ±1.61
3	Gemini 2.5 Pro Preview (June '25)	21.64% ±1.61
4	o3 (high) (April '25)	20.32% ±1.58
5	GPT-5 Mini (August '25)	19.44% ±1.55

Show all 10

Try Top 4 | Full Results

SimpleBench

Asks "trick" questions that require common-sense reasoning rather than memorized facts. Models must avoid being misled by common traps.

	MODEL	SCORE
1	Gemini 3 Pro Preview	76.4%
2	Gemini 2.5 Pro Preview (Jun '25)	62.4%
3	Claude Opus 4.5	62.0%
4	GPT-5 Pro	61.6%
5	Grok 4	60.5%

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METR Time Horizons

METR's time horizon is the human task duration at which an AI model reaches 50% success. Tasks are drawn from RE-Bench, HCAST and SWAA, which cover machine-learning research engineering, general software engineering and software operations respectively.

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	MODEL	MINUTES
1	GPT-5 (medium)	137.3 ±102.1
2	Claude Sonnet 4.5	113.3 ±91.4
3	Grok 4	110.1 ±91.8
4	Claude Opus 4.1	105.5 ±69.2
5	o3 (medium)	91.3 ±58.8

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SWE-bench Verified

A human-curated subset of 500 GitHub issues from the SWE-bench dataset tests whether models can implement valid code fixes. The model interacts with a Python repository and must modify the correct files to fix the issue. The solution is judged by running unit tests.

Show More

	MODEL	SCORE
1	Claude Sonnet 4.5	64.8% ±2.1
2	Claude Opus 4.1	63.2% ±2.2
3	Claude Opus 4	62.2% ±2.2
4	Claude Sonnet 4	60.6% ±2.2
5	Claude Haiku 4.5	60.6% ±2.2

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GPQA Diamond

A multiple-choice set of 198 PhD-level science questions in biology, chemistry and physics. It focuses on "Diamond" items for which domain experts answered correctly while non-experts often failed; random guessing yields ~25%.

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	MODEL	SCORE
1	Gemini 3 Pro Preview	92.6% ±1.7
2	Grok 4	87.0% ±2.0
3	GPT-5 (high)	86.2% ±2.1
4	GPT-5 (medium)	85.4% ±2.1
5	Gemini 2.5 Pro Preview (Jun '25)	84.8% ±2.6

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GDPval

GDPval is a new OpenAI-led benchmark spanning 44 knowledge work occupations, selected from the top 9 industries contributing to U.S. GDP, from software developers and lawyers to registered nurses and mechanical engineers. These occupations represent the types of day-to-day work where AI can meaningfully assist professionals.

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	MODEL	SCORE
1	Claude Opus 4.1	43.6%
2	GPT-5 (high)	34.8%
3	GPT-5 (medium)	33.9%
4	GPT-5 (low)	31.9%
5	o3 (high)	30.8%

Show all 9

Try Top 4 | Full Results

GSO (General Speedup Optimization)

Assesses models' ability to optimize software performance. Each task requires making code changes within a limited number of attempts. Performance is measured using OPT@K - the percentage of tasks where the model achieves at least 95% of the human-achieved speedup.

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	MODEL	SCORE
1	o3 (high)	8.8%
2	Claude Opus 4	6.9%
3	GPT-5 (high)	6.9%
4	Claude Sonnet 4	4.9%
5	Kimi K2	4.9%

DeepResearchBench

Tasks require gathering and synthesizing information from the web. Models must plan search queries and extract data from static web snapshots, covering categories like "Finding numbers", "Compiling data" and "Causal inference".

Show More

	MODEL	SCORE
1	Claude Sonnet 4.5	57.7%
2	GPT-5 (low)	57.4%
3	Claude Opus 4.1	56.4%
4	Claude Opus 4	56.3%
5	GPT-5 (medium)	56.0%

WebDev Arena

Pits two models against each other to build the best website for a given prompt. Voters rate which site looks nicer and functions better. The results are combined using the Bradley-Terry model, producing a score for each model.

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	MODEL	SCORE
1	GPT-5 (high)	1477.5
2	Claude Opus 4.1	1472.4
3	Claude Opus 4.1	1462.3
4	Claude Sonnet 4.5 (32k thinking)	1420.8
5	Gemini 2.5 Pro	1401.0

Models

BERTish

all-MiniLM-L6-v2
MPNet
MPNet-personality

↓

Feature extraction
(fine-tuning)

Open GPTs

Llama 3.2 3B

↓

Text-generation
(in-context learning)